江苏省苏州中学 国际部
Suzhou High School – International Division

SZZX-ID
Bridge Program
Physics

Unit 1 Lesson 5: free fall motion physical lab

Class Outline

Title of Unit	Linear motion
Title of Lesson	Freefall lab practical – physical lab
Starter Activity	Quiz on pre-read (5’)
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	1. How a spark timer works 2. How to measure instantaneous velocity with a spark timer 3. Safety precautions of the lab session
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
	1. Provide a worksheet for pre-read before class 2. Students pair up for experiment
	International Mindedness:
Lesson Objectives	By the end of the lesson, students will be able to: 1. Follow instructions for a lab session 2. Calculate the instantaneous velocity from looking at the average velocity 3. Record and plot data from the experiment 4. Obtain slope of a v-t graph (acceleration)
ATLs	Identify unit ATLs and how they are being addressed in the lesson
	Self-management: being punctual and meet deadlines
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute
	Inquirer
Teaching activities	Practical in a lab. Students will pair up but they will work on their own data.
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc.
	Clean up after a lab session (with a Clean Up song)

Vocabulary

- Objective 目的
- Theory 理论
- Apparatus 设备
- Procedure 过程
- Ticker-tape timer 打点计时器
- Weight 重物/砝码
- Conclusion 结论
- Mitigation 减轻措施
- Clamp 夹钳
- Analysis 分析

Ticker tape timer
(sometimes called spark timer)

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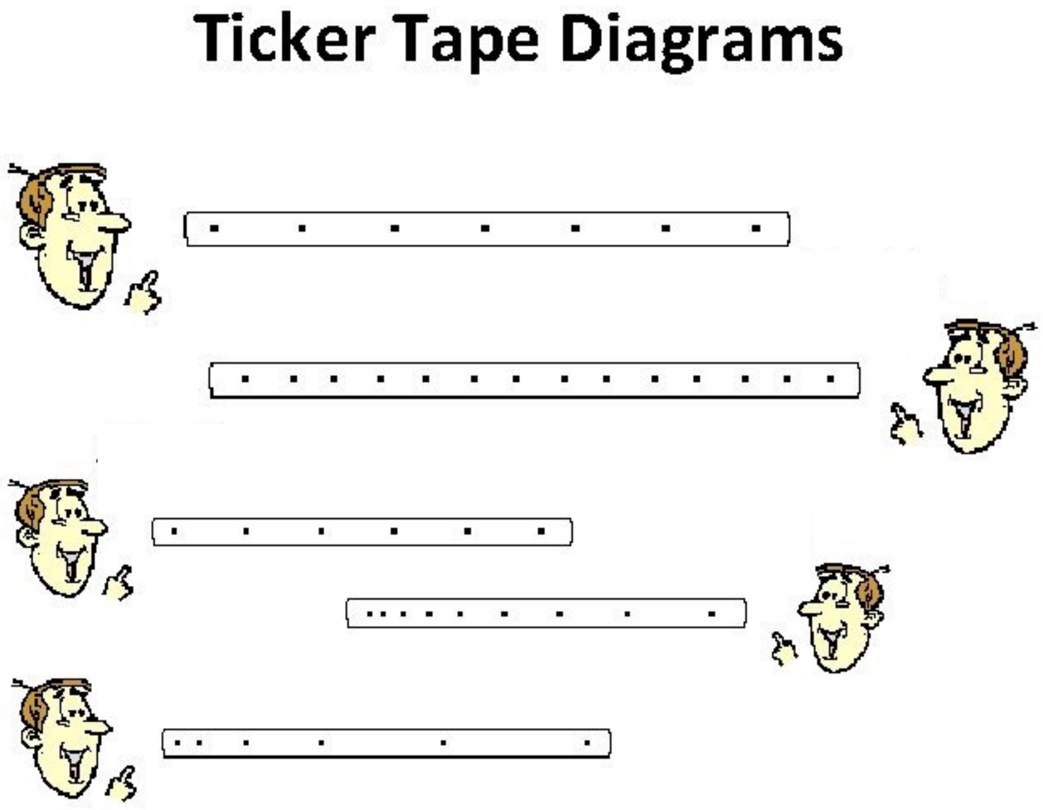
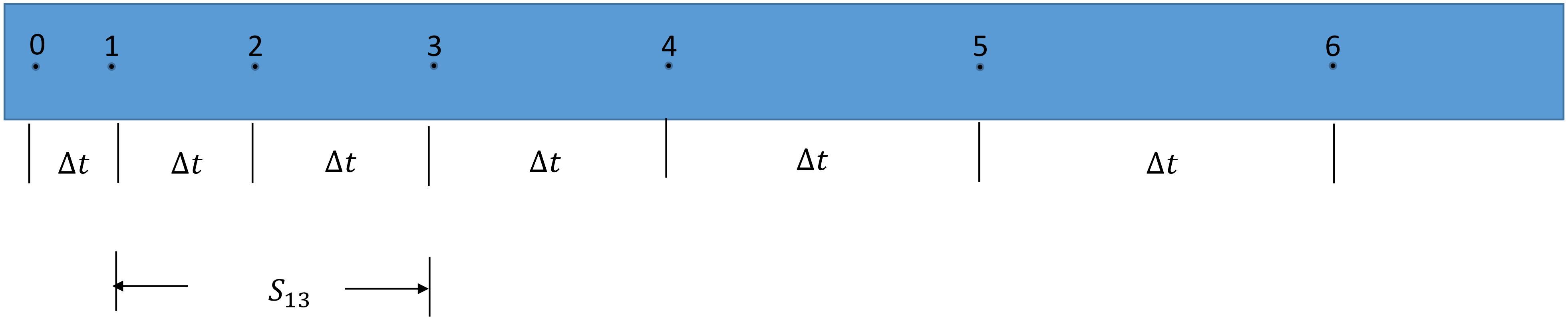


- 50 Hz ticker tape timer: make 50 dots on the tape every second
- Pattern of the tape shows the pattern of motion
- Can it measure average velocity?
- Can it measure instantaneous velocity？

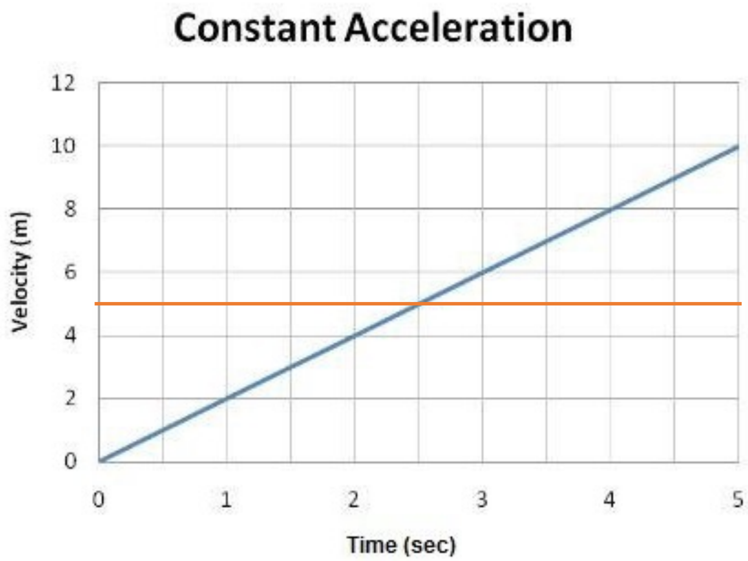
Measure average velocity from a tape



- Average velocity between dot 1 and 3

$$\bar{v} = \frac{S_{13}}{2\Delta t}$$

Measure average velocity from a tape



- In the graph at right, average velocity from 0 to 5s is 5 m/s
- It is also the instantaneous velocity at 2.5s (midpoint between 0-5s)
- On the ticker time tape, how do you find the instantaneous velocity for dot 2? What about dot 3?

Key precautions

- Connect the tape to the weight in the right way (see demonstration from your teacher)
- When inserting the tape into the ticker tape timer, make sure the tape is behind the ink wheel
- Hold the tape vertically to minimize friction (how would friction impact your result?)
- Turn on the ticker tape timer first, then release the tape

Discussion: Can you think of any safety concerns in this experiment? What to do to lower the risks?